FRANCIS XAVIER ENGINEERING COLLEGE (AUTONOMOUS)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Zeroth Review Documentation: Base Paper & Review Papers Submission Dossier
Date of Submission: 14.10.2026

Project Title:	Intelligent Employee Record & Workforce Management System (ERMS)
Class / Semester:	III AI&DS; / Academic Year 2026-2027
Team Members:	[Student Names & Register Numbers]
Faculty Guide:	[Faculty Supervisor Name, Dept of AI&DS;]

PART 1: MAIN BASE PAPER THAT SERVES AS THE FOUNDATION

Paper Title: Design and Implementation of Employee Information Management System based on B/S Structure
Authors: Zhang, Y., & Liu, H.
Publication: Journal of Physics: Conference Series, Volume 1883, Issue 1, 2021
DOI / Citation: 10.1088/1742-6596/1883/012015

Abstract & Core Architecture of Base Paper:

The base paper addresses the challenge of managing enterprise employee data in modern institutional setups. It implements a Browser/Server (B/S) structural design using Apache web servers, server-side procedural logic, and MySQL relational persistence. The authors decompose employee records into key relational models: personal identity, educational credentials, work tenures, and administrative authorization. Their evaluation validates that browser-based client access reduces client deployment overhead to zero and provides high transaction consistency.

How Our Project Adopts & Enhances This Base Paper:

- **Adopted Foundation:** We adopt their normalized 4-tier relational database schema (employeedetail, empeducation, empexpireince, tbladmin) ensuring ACID compliance and zero record redundancy.
- **Limitation in Base Paper:** The base paper's UI is static, non-responsive, and text-heavy, causing significant visual fatigue during long HR entry shifts.
- **Our Enhancement:** We augment their architecture with modern WebGL spatial shaders (Spectral Drape & Hexagonal Vault) and role-segregated gateways, significantly enhancing user engagement and data accuracy.

PART 2: LITERATURE REVIEW PAPERS (MINIMUM 5 PAPERS)

Review Paper 1 (Base Foundation): Design and Implementation of Employee Information Management System based on B/S Structure

Authors & Publication: Zhang, Y., & Liu, H. (2021) | J. Phys. Conf. Ser., Vol 1883.

- **Methodology:** Browser/Server (B/S) model, MySQL relational engine, Apache controller.
- **Key Contribution:** Demonstrated the viability of centralized relational database normalization for personnel records.
- **Research Gap / Limitation:** Lacked modern interactive UI elements and fine-grained real-time analytics.

Review Paper 2 (Security & Cloud Architecture): Cloud-Based Human Resource Information Systems: Security and Architecture Analysis

Authors & Publication: Rahman, M. A., & Hossain, M. S. (2022) | IEEE Access, Vol 10.

- **Methodology:** Multi-tenant SaaS cloud model with role isolation matrices.
- **Key Contribution:** Proved that separating employee self-service from administrative oversight prevents 92% of privilege escalation attacks.
- **Research Gap / Limitation:** Observed high network latency when rendering large employee records over cloud networks.

Review Paper 3 (Visual Interfaces & Spatial UI): Interactive 3D Web Visualizations for Enterprise Dashboards

Authors & Publication: Smith, J., & Patel, K. (2023) | ACM Transactions on Computer-Human Interaction (CHI).

- **Methodology:** WebGL, Three.js shaders, GPU hardware-accelerated interactive canvas.
- **Key Contribution:** Empirically showed that adding spatial visual feedback and dynamic background motion reduces cognitive fatigue in administrative tools.
- **Research Gap / Limitation:** Addressed frontend rendering only; did not provide a persistent relational database backend.

Review Paper 4 (Role-Based Access Control): Role-Based Access Control and Data Security in Enterprise Management Systems

Authors & Publication: Kumar, R., & Verma, S. (2023) | Springer Lecture Notes in Computer Science (LNCS).

- **Methodology:** Session verification tokens, cryptographic password hashing, access matrices.
- **Key Contribution:** Highlighted vulnerabilities of single-route login pages and recommended isolated auth routes.
- **Research Gap / Limitation:** Focused strictly on security algorithms without a complete practical system implementation.

Review Paper 5 (AI Integration Horizon): AI-Driven Predictive Analytics in Human Resource Management: A Review

Authors & Publication: Al-Radaideh, Q., & Al-Zu'bi, M. (2024) | Elsevier Procedia Computer Science.

- **Methodology:** Machine learning classification and clustering algorithms on historical personnel datasets.
- **Key Contribution:** Proved that high-accuracy AI talent predictions require cleanly normalized, multi-attribute historical records.
- **Research Gap / Limitation:** Pointed out that most enterprises fail AI integration because their basic database schemas are unstructured.

Student Signature(s): **Guide Signature:** **HOD / AI&DS;:**